

17/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

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AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

CC-144

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MM : 720

CST-6

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Physics, Chemistry and Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



PHYSICS

1. Consider the following statements and tick the correct option.

Statement A: Two cars moving on parallel tracks but in opposite directions with speed 20 m/s and 30 m/s have their relative speed 50 m/s.

Statement B: If two bodies A & B have velocities $\vec{v}_A$ and $\vec{v}_B$, then magnitude of their relative velocity $|\vec{v}_{A/B}|$ is such that $|\vec{v}_{A/B}| > |\vec{v}_A|$ and $|\vec{v}_{A/B}| > |\vec{v}_B|$.

- (1) Statement A is correct while statement B is incorrect
(2) Statement A is incorrect while statement B is correct
(3) Both statement A and statement B are correct
(4) Both statement A and statement B are incorrect

2. An atomic nucleus has mass

- (1) Less than the sum of masses of their constituent protons and neutrons
(2) Equal to the sum of masses of their constituent protons and neutrons
(3) More than the sum of masses of their constituent protons and neutrons
(4) Always equal to the total mass of constituent neutrons

3. Which of the following semiconductor is electrically neutral?

- (1) Intrinsic semiconductor
(2) p-type semiconductor
(3) n-type semiconductor
(4) All of these

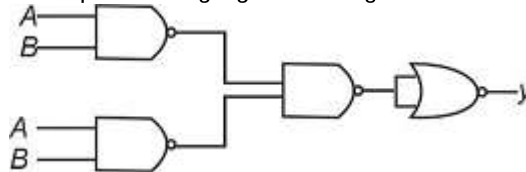
4. A hydrogen-like atom has one electron revolving around a stationary nucleus. The energy required to excite the electron from second orbit to the third orbit is 30.24 eV. The atomic number of the atom is

- (1) 2
(2) 3
(3) 4
(4) 5

5. An electron of mass m_e and a proton of mass m_p are accelerated through the same potential difference, then the ratio of de-Broglie wavelengths associated with the electron to that associated with proton is

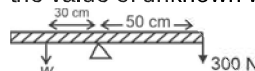
- (1) 1 : 1
(2) $\frac{m_p}{m_e}$
(3) $\frac{m_e}{m_p}$
(4) $\sqrt{\frac{m_p}{m_e}}$

6. The equivalent logic gate for the given circuit is



- (1) AND gate
(2) OR gate
(3) NOR gate
(4) NAND gate

7. If the ruler is in rotational equilibrium about the fulcrum, then the value of unknown weight (W) is



- (1) 200 N
(2) 300 N
(3) 400 N
(4) 500 N

8. The mass of a lead ball is M . If it falls down in a viscous liquid with terminal velocity v , then the terminal velocity of another lead ball of mass $4M$ in the same liquid will be

- (1) v
(2) $(4)^{1/3} v$
(3) $(16)^{1/3} v$
(4) $8v$

9. Which of the following is a correct expression for equation of continuity for liquids? (Symbols have their usual meaning)

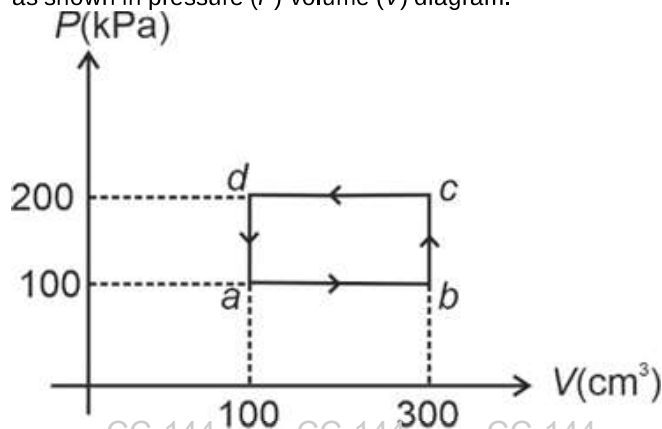
- (1) $\rho A v = \text{constant}$
(2) $\frac{A}{v} = \text{constant}$

- (3) $P + \frac{1}{2} \rho v^2 + \rho g h = \text{constant}$
(4) Both (1) and (2)

10. A system can be taken from initial state P_1, V_1 to final state P_2, V_2 by two different paths. Let Q and W represent the heat given to system and the work done by the system respectively. Which of the following must be same in both the cases?

- (1) Q
(2) W
(3) $\frac{Q}{W}$
(4) $Q - W$

11. A thermodynamic system is taken through the cycle $abcd$ as shown in pressure (P)-volume (V) diagram.



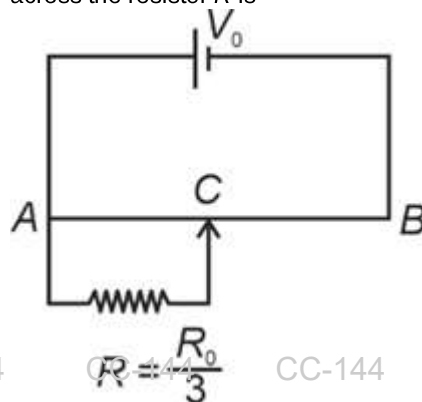
The magnitude of net heat interaction by the system in the complete cycle will be equal to

- (1) 20 J
(2) 30 J
(3) Zero
(4) 15 J
12. Consider the following statements:
(A) If pressure of an ideal gas is doubled and volume is halved, then change in its internal energy is zero.
(B) Intermolecular forces of interaction between ideal gas molecules is assumed to be zero.
(C) Rotational kinetic energy of an ideal monoatomic gas is zero.
Based on above information, pick the correct option.
- (1) Only statement (A) and (B) are correct
(2) Only statement (B) and (C) are correct
(3) Only statement (A) and (C) are correct
(4) All statements (A), (B) and (C) are correct
13. Consider the following statements:
(A) Ice point in Fahrenheit scale is 32°F
(B) Steam point in Fahrenheit scale is 252°F
Choose the correct option.
- (1) Both statements (A) and (B) are correct
(2) Both statements (A) and (B) are incorrect
(3) Statement (A) is correct while statement (B) is incorrect
(4) Statement (A) is incorrect while statement (B) is correct

14. The permittivity of free space ϵ_0 has a value of

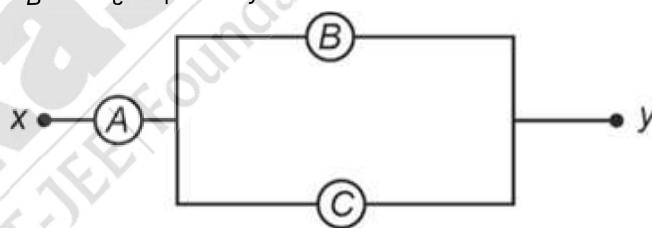
- (1) $9 \times 10^9 \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
(2) $1.6 \times 10^{19} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
(3) $8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
(4) $6.63 \times 10^{-34} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$

15. The sliding contact C is at one third of the length of a resistance wire (AB) from A as shown in the figure. If the resistance of the wire AB is R_0 , then the potential drop (V) across the resistor R is



- (1) $\frac{V_0}{2}$
(2) $\frac{V_0}{5}$
(3) $\frac{2V_0}{3}$
(4) $\frac{2V_0}{5}$

16. Three voltmeters A , B and C have resistances R , $1.5R$ and $3R$ respectively as shown in the figure. When a battery is connected between x and y , the voltmeter readings are V_A , V_B and V_C respectively. Then



- (1) $V_A \neq V_B = V_C$
(2) $V_A = V_B \neq V_C$
(3) $V_A = V_B = V_C$
(4) $V_A \neq V_B \neq V_C$

17. In a charged capacitor, the energy is associated with

- (1) The electric field between the plates
(2) The electric field outside the plates
(3) The magnetic field between plates
(4) The magnetic field outside the plates

18. If the electric flux entering and leaving an enclosed surface respectively are ϕ and 2ϕ , then the electric charge inside the surface will be

(1) $\phi\epsilon_0$
 (2) $3\phi\epsilon_0$
 (3) $2\phi\epsilon_0$
 (4) $\frac{\phi\epsilon_0}{2}$

19. Consider the following statements

(A) For a point charge, the electrostatic field varies as $\frac{1}{(\text{distance})^2}$

(B) For a short dipole, the electrostatic potential varies as $\frac{1}{r}$, where r is the distance from the centre of the dipole.

The correct statement(s) is/are

(1) Only (A)
 (2) Only (B)
 (3) Both (A) and (B)
 (4) Neither (A) nor (B)

20. A wire stretches by length l under a given load. If the load and radius of the wire are both increased to four times keeping the original length same, the stretch in the wire will be

(1) $\frac{l}{2}$
 (2) $\frac{l}{3}$
 (3) $\frac{l}{4}$
 (4) $\frac{l}{6}$

21. If the equation of a progressive wave is given by $y = 40 \sin \pi \left[\frac{t}{5} - \frac{x}{9} \right]$ where x and y are in cm and t in s, then which of the following is correct?

(1) Amplitude = 4 cm
 (2) Time period = 10 s
 (3) Speed of wave = 18 cm/s
 (4) Wavelength = 1.8 m

22. The electric and magnetic field components of an electromagnetic wave are

(1) In opposite phase and parallel to each other
 (2) In opposite phase and perpendicular to each other.
 (3) In phase and parallel to each other
 (4) In phase and perpendicular to each other

23. In YDSE, a point on the screen corresponding to the fifth minima has path difference equal to (where λ is wavelength of the light used as source)

(1) $\frac{5\lambda}{2}$
 (2) 5λ
 (3) $\frac{9\lambda}{2}$
 (4) $\frac{11\lambda}{2}$

24. A particle is under simple harmonic motion according to equation $x = A \sin \omega t$. For the given information, match the following graphs.

Column-I

Column-II

a. Position vs time

(i)

b. Velocity vs time

(ii)

c. Potential energy vs time

(iii)

d. Total energy vs time

(iv)

(1) a(i), b(ii), c(iii), d(iv)
 (2) a(ii), b(iii), c(iv), d(i)
 (3) a(ii), b(i), c(iv), d(iii)
 (4) a(iii), b(ii), c(i), d(iv)

25. A thin copper wire of length L increases in length by 4% when heated from T_1 to T_2 . If a copper cube having side L is heated from T_1 to T_2 , then the percentage change in volume of the cube is

(1) 6%
 (2) 12%
 (3) 13%
 (4) 15%

26. The value of 1 J energy in a system of units where 1 unit of mass is equal to 100 g, 1 unit of length is 10 cm and 1 unit time is equal to 5 seconds, will be

(1) 5000
 (2) 25000
 (3) 10000
 (4) 15000

27. Read the following statements and select the correct option.

Assertion (A) : If initial velocity of a particle is zero and acceleration is constant, then its path must be a straight line.

Reason (R) : If initial velocity of a particle is non-zero and acceleration is constant, then path of particle may be straight line or parabola.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true, (R) is false
- (4) (A) is false, (R) is true

28. If position of a particle that is moving along x-axis is given by $x = (2 - 3t^2 + 6t)$ m, then the ratio of displacement to the distance covered by the particle in first 2 seconds is

- (1) Zero
- (2) 6 : 1
- (3) 1 : 1
- (4) 1 : 6

29. A bomb initially at rest explodes into three fragments. Total linear momentum of all the fragments just after explosion

- (1) Depends on mass of the fragments
- (2) Depends on speeds of the fragments
- (3) Is zero
- (4) Is infinity

30. A car of weight 1000 N is moving at a constant speed on a hilly road. The radius of curvature of sections at A and B are 100 m. As the car passes over A, the normal force on the car is 100 N. The normal force on the car as it passes through the point B is ($g = 10 \text{ m/s}^2$)



- (1) 1000 N
- (2) 2000 N
- (3) 1900 N
- (4) 4000 N

31. If potential energy (U) of a particle varies with position of particle along xy-plane as $U = x^3 - x^2 + 2y + y^2$. The force acting on the particle at a position $P(1, 2)$ is (all quantities are in SI unit)

- (1) $-(\hat{i} + 6\hat{j}) \text{ N}$
- (2) $(6\hat{i} + \hat{j}) \text{ N}$
- (3) $(\hat{i} + \hat{j}) \text{ N}$
- (4) Zero

32. An elevator motor produces 2400 W power. The constant speed with which it can lift a 800 kg load is

- (1) 0.6 m/s
- (2) 0.3 m/s
- (3) 2 m/s
- (4) 1 m/s

33. The position vector of two particles each of mass 1 kg, (A) and (B) are given by $\vec{r}_A = (\hat{i} + 2\hat{j}) \text{ m}$ and $\vec{r}_B = (2\hat{i} - \hat{j}) \text{ m}$ respectively. Velocity of A is $(4\hat{k}) \text{ m/s}$ while particle B is at rest. Angular momentum of particle (A) with respect to the position of particle (B) is (in $\text{kg m}^2 \text{ s}^{-1}$)

- (1) $\hat{i} + 3\hat{j}$
- (2) $\hat{i} - 3\hat{j}$
- (3) $12\hat{i} + 4\hat{j}$
- (4) $4\hat{i} + \hat{j}$

34. Given below are two statements:

Statement I: The force of attraction between a solid spherical body of uniform mass density and a point mass m situated inside is just as if the entire mass of the solid body is concentrated at its centre.

Statement II: Force on a point mass m situated inside a spherical shell is zero.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct while statement II is incorrect
- (4) Statement I is incorrect while statement II is correct

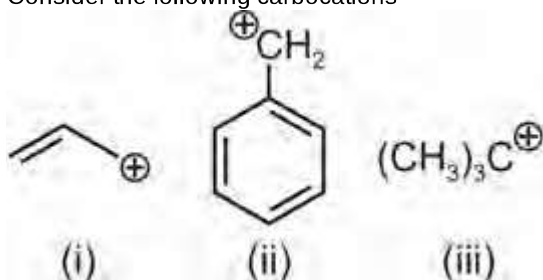
35. Time period of a satellite orbiting very close to the surface of earth is nearly (assume radius of earth $R_e = 6400 \text{ km}$)

- (1) 84.6 minutes
- (2) 30 minutes
- (3) 120 minutes
- (4) 160 minutes

36. If all the following bodies have same mass and same radius, then the one having least moment of inertia is
- (1) A hollow sphere about its tangent
 - (2) A ring about its axis
 - (3) A disc about its axis
 - (4) A solid sphere about its diameter
37. The distance from the object to the principal focus of a concave mirror is x . If focal length of mirror is f , the distance of image from focus is
- (1) $\frac{f(f+x)}{x}$
 - (2) $\frac{f^2}{x}$
 - (3) $\frac{x|x-f|}{f}$
 - (4) $\frac{x^2}{f}$
38. For a real object, magnification produced by a convex mirror is always
- (1) Less than one
 - (2) Greater than one
 - (3) Equal to one
 - (4) Less than or greater than one
39. The total number of magnetic field lines originating or terminating on a pole of strength ' m ' is proportional to
- (1) $\mu_0 m^2$
 - (2) $\frac{\sqrt{m}}{\mu_0}$
 - (3) $\mu_0 m$
 - (4) m^2
40. A long solid copper pipe carrying current uniformly across its area, the magnetic field produced will be
- (1) Inside the pipe only
 - (2) Outside the pipe only
 - (3) Neither inside nor outside the pipe
 - (4) Both inside and outside the pipe
41. A circular coil of radius 20 cm, carries current of 5 A. If it has 28 turns, then the magnetic field at the centre of the coil is (in Wb/m^2)
- (1) 4.4×10^{-4}
 - (2) 14×10^{-3}
 - (3) 2×10^{-3}
 - (4) 2.2×10^{-5}
42. If instantaneous current is given by $I = (3\sqrt{2} + 8 \sin \omega t)$ A, then effective value of current in the circuit is
- (1) 5 A
 - (2) $5\sqrt{2}$ A
 - (3) 10 A
 - (4) $10\sqrt{2}$ A
43. Read the following statements and select the correct option.
Statement A : In an AC circuit, the resistance of a resistor does not depend upon frequency of the source.
Statement B : Capacitive reactance of capacitor for a d.c. source is infinite.
- (1) Both statement A and statement B are correct
 - (2) Both statement A and statement B are incorrect
 - (3) Statement A is correct while statement B is incorrect
 - (4) Statement A is incorrect while statement B is correct
44. Magnetic flux through a coil of resistance 10Ω is changed from 10 Wb to 50 Wb in 2 ms. Charge passed through the coil during this time is
- (1) 2 C
 - (2) 4 C
 - (3) 3 C
 - (4) 6 C
45. The magnetic energy stored in an inductor of inductance 4 μH is 8 μJ . The current flowing through the inductor will be
- (1) 1 A
 - (2) 3 A
 - (3) 2 A
 - (4) 4 A

CHEMISTRY

46. Consider the following carbocations



The correct order of stability is

- (1) (iii) > (i) > (ii)
- (2) (iii) > (ii) > (i)
- (3) (i) > (ii) > (iii)
- (4) (iii) > (i) > (iii)

47. Consider the following statements

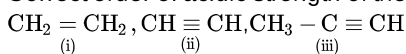
- (a) Cellulose is a straight chain polysaccharide composed only of β -D-glucose units.
 - (b) The carbohydrates are stored in animal body as glycogen.
 - (c) Glycogen is found in yeast.
- The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

48. If a 15 watt bulb emits monochromatic light of wavelength 528 nm then the number of photons emitted per second by the bulb will be nearly ($h = 6.6 \times 10^{-34}$ Js)

- (1) 2×10^{21}
- (2) 4×10^{19}
- (3) 8×10^{18}
- (4) 2.5×10^{20}

49. Correct order of acidic strength of the given compounds is



- (1) (ii) > (i) > (iii)
- (2) (i) > (ii) > (i)
- (3) (iii) > (ii) > (i)
- (4) (ii) > (iii) > (i)

50. Given below are the two statements

Statement I: Tryptophan is an essential amino acid.
Statement II: All naturally occurring α -amino acids are optically active compounds.
In light of above statements, choose the correct answer

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

51. Match List-I with List-II

	List-I (Ion)		List-II (Number of unpaired electrons)
a.	Mn^{3+}	(i)	5
b.	Cr^{4+}	(ii)	3
c.	Ni^{3+}	(iii)	2
d.	Co^{4+}	(iv)	4

The correct match is

- (1) a(iii), b(iv), c(ii), d(i)
- (2) a(iv), b(iii), c(i), d(ii)
- (3) a(ii), b(iii), c(i), d(iv)
- (4) a(iv), b(iii), c(ii), d(i)

52. Given below are two statements

Statement I: Inversion of cane sugar in acidic medium is pseudo first order reaction.

Statement II: Rate of reaction during inversion of cane sugar in acidic medium depends on the concentration of cane sugar.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

53. For a zero order reaction half-life will be (a_0 = initial concentration)

- (1) $\frac{0.693}{k}$
- (2) $\frac{a_0}{k}$
- (3) $\frac{2a_0}{k}$
- (4) $\frac{a_0}{2k}$

54. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)

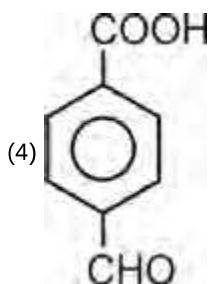
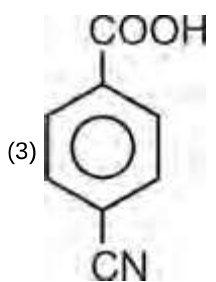
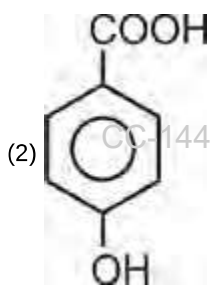
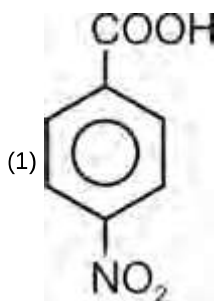
Assertion (A): 2-Methyl propanal will not give aldol condensation reaction to produce an enal.

Reason (R): 2-Methyl propanal does not contain α -hydrogen.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

55. Most acidic compound among the following is



56. Color of $[\text{Fe}(\text{SCN})]^{2+}$ will be

- (1) Yellow
- (2) Blue
- (3) Black
- (4) Blood red

57. Match cations in List-I with their group reagent in List-II

	List-I		List-II
a.	Ba^{2+}	(i)	$\text{H}_2\text{S} + \text{dil. HCl}$
b.	Cu^{2+}	(ii)	$\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
c.	Al^{3+}	(iii)	$\text{H}_2\text{S} + \text{NH}_4\text{OH}$
d.	Ni^{2+}	(iv)	$(\text{NH}_4)_2\text{CO}_3 + \text{NH}_4\text{OH}$

Choose the correct answer from the options given below.

(1) a(ii), b(iii), c(i), d(iv)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(iv), b(ii), c(i), d(iii)

58. Match List-I with List-II

List-I	List-II
[Co(NH ₃) ₆] [Cr(CN) ₆]	
a. and [Cr(NH ₃) ₆] [Co(CN) ₆]	(i) Solvate isomers
[Co(NH ₃) ₅ SO ₄]Br	
b. and [Co(NH ₃) ₅ Br]SO ₄	(ii) Linkage isomers
[Co(NH ₃) ₅ (NO ₂)]Cl ₂	
c. and [Co(NH ₃) ₅ ONO]Cl ₂	(iii) Coordination isomers
[Cr(H ₂ O) ₆]Cl ₃	
d. and [Cr(H ₂ O) ₅ Cl]Cl ₂ ·H ₂ O	(iv) Ionisation isomers

Choose the correct answer from the options given below

(1) a(i), b(ii), c(iv), d(iii)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(iii), b(iv), c(ii), d(i)

59. Which of the following is an inner orbital complex?

(1) $[\text{Co}(\text{NH}_3)_6]^{3+}$

(2) $[\text{Ni}(\text{NH}_3)_6]^{2+}$

(3) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

(4) $[\text{FeF}_6]^{3-}$

60. Given below are two statements

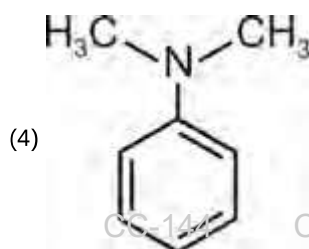
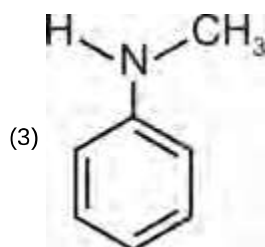
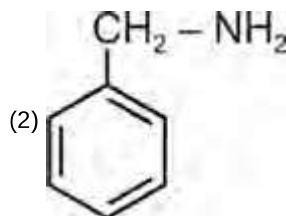
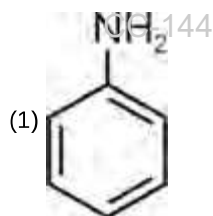
Statement I : Aniline reacts with bromine water at room temperature to give a white ppt of 2,4,6-tribromoaniline.

Statement II : Aniline does not undergo Friedel-Crafts reaction.

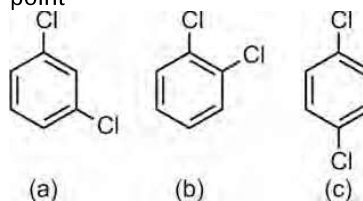
In the light of above statements, choose the most appropriate answer from the options given below

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

61. Which of the following amines can be prepared by Gabriel phthalimide synthesis?



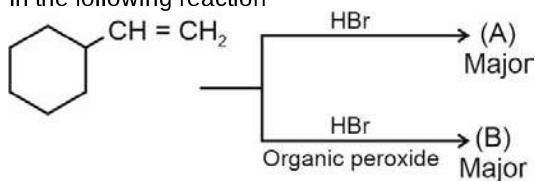
62. Arrange the following in their decreasing order of melting point



- (1) (b) > (c) > (a)
- (2) (c) > (a) > (b)
- (3) (c) > (b) > (a)
- (4) (a) > (b) > (c)


Aakash
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63. In the following reaction



Products (A) and (B) respectively are

- (1) and
- (2) and
- (3) and
- (4) and

64. Given below are two statements

Statement I: The stable form of sulphur at room temperature is rhombic sulphur, which transforms to monoclinic sulphur when heated above 369 K.

Statement II: Rhombic sulphur crystals are formed on evaporating the solution of roll sulphur in CS_2 .

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both statement I and statement II are correct
 (2) Statement I is correct but statement II is incorrect
 (3) Both statement I and statement II are incorrect
 (4) Statement I is incorrect but statement II is correct

65. A Xenon fluoride compound (X) upon partial hydrolysis gives XeO_2F_2 . The number of lone pair of electrons present in compound (X) is

- (1) 1
 (2) 18
 (3) 19
 (4) 12

66. The pair that represent the neutral oxides of nitrogen is

- (1) N_2O and NO
 (2) N_2O and N_2O_3
 (3) N_2O_3 and N_2O_4
 (4) N_2O_5 and NO

67. Match the List-I with List-II

List-I (Reaction)		List-II (Major product)	
a.	$\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{OH} \xrightarrow[573\text{ K}]{\text{Cu}} \text{CH}_3-\text{C}(\text{CH}_3)=\text{CH}_2$	(i)	Salicylic acid
b.	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{OH} \xrightarrow{\text{PCC}}$	(ii)	2-methylpropene
c.	$\text{C}_6\text{H}_5\text{COOH} \xrightarrow{(\text{CH}_3\text{CO})_2\text{O}, \text{H}^+}$	(iii)	But-2-enal
d.	$\text{C}_6\text{H}_5\text{OH} \xrightarrow[\text{(iii) H}_3\text{O}^+]{\text{(i) NaOH, (ii) CO}_2}$	(iv)	Aspirin

Choose the correct answer from the options given below :

- (1) a(ii), b(iii), c(iv), d(i)
 (2) a(ii), b(iii), c(i), d(iv)
 (3) a(iii), b(i), c(iv), d(ii)
 (4) a(iii), b(ii), c(i), d(iv)

68. Oxidation of phenol with chromic acid produces

- (1) Benzaldehyde
- (2) Benzoquinone
- (3) Resorcinol
- (4) Hydroquinone

69. 3-methylbut-1-ene when subjected to hydroboration oxidation, then the product formed is

- (1) 3-methylbutan-2-ol
- (2) 2-methylbutan-2-ol
- (3) 3-methylbutan-1-ol
- (4) Pentan-2-ol

70.

	List-I (Species)		List-II (Bond order)
a.	O_2	(i)	1.5
b.	O_2^-	(ii)	2
c.	N_2	(iii)	2.5
d.	N_2^+	(iv)	3

The correct match is

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(ii), b(i), c(iv), d(iii)
- (3) a(iv), b(iii), c(ii), d(i)
- (4) a(iii), b(iv), c(i), d(ii)

71. Given below are two statements

Statement I: Resonance stabilizes the molecule as the energy of the resonance hybrid is less than the energy of any single canonical structure.

Statement II: There is equilibrium between the canonical forms.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Both statement I and statement II are incorrect
- (4) Statement I is incorrect but statement II is correct

72. Formal charge on C in CO_3^{2-} is

- (1) -4
- (2) 0
- (3) -2
- (4) 4

73. The mass of urea (NH_2CONH_2) required to make 3 kg of 0.3 molal aqueous solution is

- (1) 54 g
- (2) 53.04 g
- (3) 27 g
- (4) 26.52 g

74. Given below are two statement one is labelled as **Assertion (A)** and other is labelled **Reason (R)**.

Assertion (A) : Solubility of oxygen in water decreases with rise in temperature

Reason (R) : Value of Henry's law constant for O_2 decreases with increase in temperature.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

75. If the concentration of NaOH aqueous solution is 4 gL^{-1} then the volume of HCl gas at STP required to neutralize 20 mL of the same NaOH solution will be

- (1) 22.4 mL
- (2) 44.8 mL
- (3) 11.2 mL
- (4) 224 mL

76. Given below are two statements :

Statement I : Both La^{3+} and Lu^{3+} ions are colourless.

Statement II : Ce^{4+} is paramagnetic whereas Yb^{2+} is diamagnetic.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Both statement I and statement II are incorrect
- (4) Statement I is incorrect but statement II is correct

77. The correct order of negative enthalpy of hydration for the following species.

- (1) $Ni^{2+} > Co^{2+} > Fe^{2+} > Ti^{2+}$
- (2) $Ti^{2+} > Fe^{2+} > Co^{2+} > Ni^{2+}$
- (3) $Co^{2+} > Ni^{2+} > Fe^{2+} > Ti^{2+}$
- (4) $Ni^{2+} > Fe^{2+} > Ti^{2+} > Co^{2+}$

78. The species which by definition has zero standard molar enthalpy of formation at 298 K is

- (1) $\text{C}_2\text{H}_6(\text{g})$
- (2) $\text{H}_2\text{O}(\text{g})$
- (3) $\text{Cl}_2(\text{g})$
- (4) $\text{Br}_2(\text{g})$

79. The standard enthalpies of formation of $\text{CO}_2(\text{g})$, $\text{H}_2\text{O}(\text{l})$ and glucose (s) at 298 K are -400 kJ/mol , -300 kJ/mol and -1300 kJ/mol , respectively. The standard enthalpy of combustion per mole of glucose at 298 K is

- (1) -2900 kJ
- (2) 2800 kJ
- (3) -16.11 kJ
- (4) -2800 kJ

80. Match column-I with column-II

	Column-I (salts)		Column-II (pH)
a.	0.1 M NH_4Cl	(i)	$\text{pH} = \frac{1}{2}$ $(\text{pK}_w + \text{pK}_a + \log C)$
b.	0.1 M NaCl	(ii)	$\text{pH} = \frac{1}{2}$ $(\text{pK}_w - \text{pK}_b - \log C)$
c.	0.1 M $\text{CH}_3\text{COONH}_4$	(iii)	$\text{pH} = \frac{\text{pK}_w}{2}$
d.	0.1 M CH_3COONa	(iv)	$\text{pH} = \frac{1}{2}$ $(\text{pK}_w + \text{pK}_a - \text{pK}_b)$

Choose the correct answer from the options given below.

- (1) a(i), b(iii), c(ii), d(iv)
- (2) a(ii), b(i), c(iv), d(iii)
- (3) a(ii), b(iii), c(iv), d(i)
- (4) a(iv), b(iii), c(ii), d(i)

81. The solubility of Ag_2CrO_4 in 0.2 M AgNO_3 aqueous solution is (in mol/L). [Given that K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K]

- (1) 3.26×10^{-11}
- (2) 2.75×10^{-10}
- (3) 2.75×10^{-11}
- (4) 3.26×10^{-10}

82. The correct order of negative electron gain enthalpy of halogen is

- (1) $\text{Cl} > \text{F} > \text{I} > \text{Br}$
- (2) $\text{I} > \text{Cl} > \text{Br} > \text{F}$
- (3) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
- (4) $\text{F} > \text{Cl} > \text{Br} > \text{I}$

83. Symbol of an element having atomic number 109 is

- (1) Uen
- (2) Uep
- (3) Une
- (4) Unn

84. The molar conductivity of 0.2 M solution of NaCl having conductivity of $1.25 \times 10^{-3} \text{ S cm}^{-1}$ is

- (1) $0.625 \text{ S cm}^2 \text{ mol}^{-1}$
- (2) $5.0 \text{ S cm}^2 \text{ mol}^{-1}$
- (3) $6.25 \text{ S cm}^2 \text{ mol}^{-1}$
- (4) $1.25 \text{ S cm}^2 \text{ mol}^{-1}$

85. A current of 4.825 ampere flowing in an electrolytic cell for 100 seconds, 0.15 g of metal deposits at electrode, the equivalent weight of metal is

- (1) 10
- (2) 30
- (3) 45
- (4) 15

86. Consider the following statements :

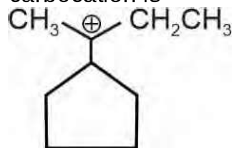
Statement I: The carbon compound formed on complete combustion of ethyne has the same hybridisation of carbon as that of the carbon in ethyne.

Statement II: Ethyne molecule has one σ bond and two pi bonds in its structure.

In the light of above statements, choose the correct option.

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Both statement I and statement II are incorrect
- (4) Statement I is incorrect but statement II is correct

87. The total number of contributing structures showing hyperconjugation (involving C – H bonds) for the following carbocation is



- (1) 7
(2) 6
(3) 5
(4) 9
88. Choose the incorrect statements :
- (1) Methoxyethane does not show metamerism.
(2) Propanal and propanone are functional group isomers
(3) 2-methylbutane and n-pentane are chain isomers
(4) 2,2-Dimethylpropane and 2,3-Dimethylbutane are position isomers

89. Which of the following oxide is amphoteric in nature?

- (1) CO₂
(2) CO
(3) SnO₂
(4) GeO₂

90. Total number of 3 centre 2-electron bonds in one molecule of diborane is

- (1) 1
(2) 2
(3) 3
(4) 4

BIOLOGY

91. Select the **incorrectly** matched pair.

- (1) Primary consumers – Also known as key industry animals
(2) Primary producers – Trap 50% energy of PAR
(3) Secondary consumers – Feed on herbivores
(4) Tertiary consumers - Generally occupy top most trophic level in ecological pyramid

92. Select the processes in which decarboxylation reaction takes place.

- (a) Krebs' cycle
(b) Link reaction
(c) Glycolysis
(d) Lactic acid fermentation
(e) Alcoholic fermentation

The **correct** ones are

- (1) (a), (b) and (d)
(2) (b), (c) and (d)
(3) (a) and (e) only
(4) (a), (b) and (e)

93. Select the option for all the correct statements w.r.t. chemiosmotic hypothesis.

- (a) For ATP synthesis, protons accumulate towards the stroma of the thylakoid
(b) Splitting of the water molecule takes place on the inner side of the thylakoid membrane
(c) Primary acceptor of electron is located towards the inner side of the inner membrane of chloroplast
(d) NADP reductase enzyme is located on the stroma side of the thylakoid membrane

- (1) (a) and (b)
(2) (c) and (d)
(3) (b) and (d)
(4) (a) and (c)

94. In sorghum plant, the first stable product of primary CO₂ fixation is

- (1) Pyruvic acid
(2) Oxaloacetic acid
(3) Phosphoglyceric acid
(4) Succinic acid

95. What is the net gain of ATP when a molecule of glucose is converted to two molecules of pyruvic acid?
- Four
 - Six
 - Two
 - Eight
96. Vertical distribution of different species occupying different levels in an ecosystem is called
- Species composition
 - Speciation
 - Energy flow
 - Stratification
97. Which of the following statements is **not** correct w.r.t. characteristics of living beings?
- Growth involves both increase in mass and number of
- cells which takes place from inside the body of living organism and is irreversible.
 - In higher plants and animals, growth and reproduction are linked but are mutually exclusive events.
 - Sensitivity or awareness is regarded as defining property of living organism.
 - Reproduction is one of the fundamental characteristics of living organisms and can be regarded as defining characteristic of all living organisms
98. The sole members of Kingdom Monera have
- Cell wall composed of peptidoglycan
 - True nucleus
 - 60S and 40S subunits of ribosome
 - Membrane bound cell organelles
99. In all of the given fungi, only asexual or vegetative phases are known, **except**
- Alternaria*
 - Trichoderma*
 - Colletotrichum*
 - Rhizopus*
100. Identify the **mismatched** pair w.r.t. symbols used in floral formula.
- Polypetalous – C_n
 - Epipetalous stamens – $\overline{P A}$
 - Epipetalous stamens – $\overline{C A}$
 - Gamopetalous – $K_{(n)}$
101. Consider the following given characters.
- Flowers are bisexual usually actinomorphic but sometimes zygomorphic.
 - Bicarpellary, syncarpous, superior ovary, unilocular but sometimes bilocular due to false septum.
 - Seeds are small and non-endospermic.
 - Silique type of fruit is present.
- The above given floral characters are **true** for members of the family
- Fabaceae
 - Solanaceae
 - Brassicaceae
 - Gramineae
102. Read the following statements and select the **correct** option.
- Statement A:** In maize seeds the outer covering of endosperm separates the embryo by a proteinaceous layer called aleurone layer.
- Statement B:** Remains of nucellus persist in the seeds of some plants and is called perisperm.
- Only statement A is correct
 - Only statement B is correct
 - Both statements A and B are correct
 - Both statements A and B are incorrect
103. Read the following assertion(A) and reason(R) and select the **correct** option.
- Assertion(A):** In racemose type of inflorescence, the main axis continues to grow.
- Reason(R):** In racemose inflorescence, the younger flowers are present towards the apex and the older ones are at the base.
- (A) is true but (R) is false
 - Both (A) and (R) are false
 - Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true but (R) is not the correct explanation of (A)
104. Which of the following cell organelles is associated with detoxification of drugs?
- Rough endoplasmic reticulum
 - Smooth endoplasmic reticulum
 - Golgi body
 - Lysosome
105. Which of the following statements **correctly** describes the function of nucleolus?
- It is rich in hydrolytic enzyme which are capable of digesting carbohydrates, proteins etc.
 - It is involved in the movement of substances in and out of the nucleus
 - It is the site for ribosomal RNA (rRNA) synthesis
 - It helps in osmoregulation and excretion in *Amoeba*

106. Complete the analogy by selecting the **correct** option.

Cuboidal epithelium : PCT of nephron :: Squamous epithelium : _____

- (1) Bronchioles
- (2) Blood vessels
- (3) Pancreatic ducts
- (4) Small intestine

107. Consider the following:

- (a) Non-polymeric in nature
 - (b) Has 16 carbons including the carboxyl carbon
 - (c) Belongs to the category of saturated fatty acid
- The above given features hold **true** for which of the following molecules?

- (1) Arachidonic acid
- (2) Glutamic acid
- (3) Cytidylic acid
- (4) Palmitic acid

108. Ribose differs from glucose in all of the following aspects, **except**

- (1) Number of carbon atoms
- (2) Being a component of nucleic acid
- (3) Being a monosaccharide
- (4) Number of hydroxyl groups in the structure

109. Choose the neutral amino acid among the following which is also considered as the simplest amino acid.

- (1) Valine
- (2) Glycine
- (3) Proline
- (4) Tyrosine

110. Select the option representing the molecules having peptide bond and glycosidic bond respectively.

- (1) Lecithin, Glucagon
- (2) Insulin, Glycogen
- (3) Inulin, Insulin
- (4) Myoglobin, Collagen

111. Carbonic anhydrase shares the same co-factor as that of carboxypeptidase. Select the **correct** option w.r.t. this co-factor.

- (1) Haem
- (2) Zinc
- (3) Magnesium
- (4) Chloride

112. Select the **incorrect** statement among the following.

- (1) Most of the cartilages in vertebrate embryos are replaced by bones in adults.
- (2) The bone marrow in all the bones in animals is the site of production of blood cells.
- (3) The bone cells (osteocytes) are present in the spaces called lacunae.
- (4) Limb bones interact with skeletal muscles attached to them to bring about movements.

113. Amongst all the organisms mentioned below, choose the **one** that contains a network of neurons but no brain.

- (1) *Hippocampus*
- (2) *Canis*
- (3) *Hydra*
- (4) *Anopheles*

114. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Transformation	(i)	Complete set of genetic information in an organism
b.	Spooling	(ii)	Extraction of DNA from DNA bands cut out from agarose gel
c.	Elution	(iii)	Introduction of foreign DNA in the host bacterium
d.	Genome	(iv)	Method of obtaining DNA in a form of spool over a glass rod

- (1) a(ii), b(i), c(iii), d(iv)
- (2) a(iii), b(ii), c(iv), d(i)
- (3) a(iii), b(iv), c(ii), d(i)
- (4) a(ii), b(iv), c(i), d(iii)

115. If one wants to generate a high copy number of a desired DNA segment, then he/she should take a vector which

- (1) Has many restriction sites for a single restriction enzyme
- (2) Is bigger in size
- (3) Has selectable markers controlling the copy number
- (4) Has the *ori* site to support high copy number

116. Most of the cell organelles duplicate during

- (1) Post mitotic phase
- (2) DNA synthesis phase
- (3) Pre-mitotic phase, i.e., G₂ phase
- (4) M-phase

117. Which of the following are exclusive events/phases of meiosis, if we compare it with mitosis?

- Formation of synaptonemal complex
- Separation of chromatids in anaphase
- Formation of recombination nodule
- Intrameiotic interphase

The **correct** ones are

- b and c only
- a, c and d only
- a and d only
- a, b and c only

118. Read the following statements and state them true (T) or false (F) and choose the **correct** option.

- Cytoskeleton helps in mechanical support and cell motility.
- Inner membrane of the mitochondria has enzymes and electron carriers.
- Secondary cell wall is added internally to the primary cell wall and diminishes with maturity.
- Golgi apparatus is responsible for processing and packaging of lipids and proteins.

	(A)	(B)	(C)	(D)
(1)	T	F	F	T
(2)	T	T	F	T
(3)	T	F	T	T
(4)	F	F	T	F

- (1)
- (2)
- (3)
- (4)

119. Stage which shows transition to metaphase-I is also associated with

- Tetrad formation
- Dissolution of synaptonemal complex
- Terminalisation of chiasmata
- Condensation of chromosomal material

120. Which among the following plays structural and catalytic role during translation?

- Sn RNA
- rRNA
- tRNA
- mRNA

121. Select the **mismatched** pair.

- Minisatellites – Variable number of Tandem repeats
- Frederick Sanger – Developed method for determination of amino acid sequences in proteins
- Dystrophin gene – Largest known human gene
- George Gamow – Cell free system for protein synthesis helped in deciphering the genetic code

122. Match List-I with List-II.

	List-I		List-II
a.	Severo Ochoa enzyme	(i)	Catalyses the addition of methyl guanosine triphosphate at 5' end of hnRNA
b.	Helicase	(ii)	Join the two discontinuously synthesised fragments of DNA
c.	Guanylyl transferase	(iii)	Facilitates unwinding of double helix of DNA during replication
d.	DNA Ligase	(iv)	Helpful in polymerising RNA with defined sequences

Choose the **correct** option.

- a(iii), b(i), c(ii), d(iv)
- a(ii), b(iv), c(iii), d(i)
- a(iv), b(iii), c(i), d(ii)
- a(iv), b(i), c(ii), d(iii)

123. Xylem and phloem within the vascular bundle are arranged in an alternate manner along different radii. Such arrangement is found in

- Both leaves and roots of monocots
- Roots of angiosperms
- Both stems and roots of dicots
- Leaves and stems of all angiosperms

124. Read the given statements (A-D) and choose the option with **incorrect** one(s).

- Roots of grasses differ from roots of sunflower, as the former lack suberin depositions in the innermost layer of cortex.
- In a typical young dicot stem, the outermost protective layer may bear trichomes and a few stomata.
- Ground tissue system forms the main bulk of the plant body.
- Monocotyledonous and dicotyledonous plants show marked variations in their internal structures but they do not differ in type and location of vascular bundles.

- B and C only
- All A, B, C and D
- A and D only
- D only

125. DNA fingerprinting is useful in

- Paternity testing in case of disputes
- Forensics
- Determining population and genetic diversities
- Creation of new genetic material to cure genetic disorders

The **correct** one(s) is/are

- b and d only
- c only
- a and b only
- a, b and c

126. Tracheids show similarity with xylem parenchyma as they both are

- Components of complex permanent tissue
- Devoid of protoplasm
- Interconnected through perforations in their common walls
- Involved in storage of starch, fats and tannins

127. Of the total, nitrogenous bases in a fragment of dsDNA, guanine was found to be 46%. Thymine and cytosine in this fragment would respectively be

- 4% and 92%
- 4% and 46%
- 8% and 4%
- 8% and 46%

128. Which of the following wall layers of anther is responsible to provide nourishment to the developing pollen grains?

- Tapetum
- Endothecium
- Middle layer
- Epidermis

129. In roots, the cells proximal to the meristematic zone towards the base of stem do **not**

- Represent the phase of elongation
- Involve in elongation of root in length
- Protect the tender apex of the root
- Involve cell enlargement for growth

130. There are leaf A and leaf B having the initial areas of 10 cm^2 and 25 cm^2 , respectively. Leaf A increased its area by 20 cm^2 and leaf B by 10 cm^2 in a unit time. What are their relative growth rates?

Choose the **correct** option from the following.

- A-50% ; B-25%
- A-200% ; B- 40%
- A-100% ; B- 25%
- A-50% ; B- 40%

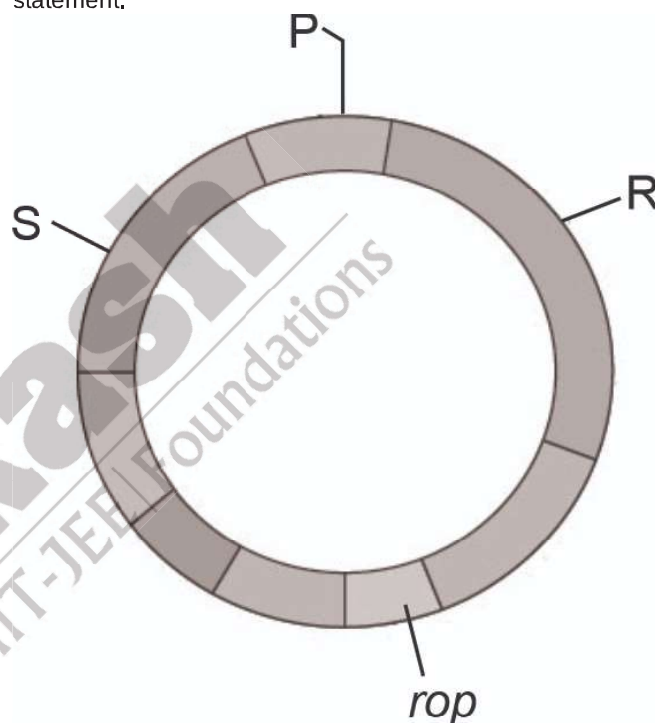
131. During agarose gel electrophoresis, DNA moves across the gel, due to the sieving effect provided by the

- Negatively charged DNA molecule
- Electric field generated by the electrodes
- Positively charged DNA molecules
- Agarose gel

132. Which of the following bonds get broken during the denaturation step of PCR?

- Peptide
- Hydrogen
- Glycosidic
- Ester

133. Consider the given figure of pBR322 and select the correct statement.



- 'P' has the site for the first discovered restriction enzyme.
- 'R' codes for the enzyme that degrades the ampicillin.
- rop* is involved in the replication of the plasmid.
- 'S' has the site for origin of replication.

134. A part of human brain called 'X' has a centre which controls urge for eating. Identify 'X' and choose the **incorrect** option w.r.t it.

- It is the basal part of the diencephalon.
- The hormones released by it regulate the synthesis and secretion of anterior pituitary hormones.
- Along with the limbic system, it regulates the expression of emotional reactions.
- It acts as the major coordinating centre for sensory and motor signalling.

135. Approximate number of seminiferous tubules in each testis of a healthy adult man is

- (1) 750 - 1000
- (2) 750 - 1500
- (3) 1 - 3
- (4) 250 - 750

136. Which of the following is a part of the human female reproductive system but is not categorised as the external genitalia?

- (1) Labia minora
- (2) Mons pubis
- (3) Clitoris
- (4) Vagina

137. Select the unpaired structure present in the brain of *Rana tigrina*.

- (1) Olfactory lobe
- (2) Optic lobe
- (3) Diencephalon
- (4) Cerebral hemisphere

138. Comprehend the following features w.r.t. humans.

- (i) Connected to the pelvic wall and uterus by ligaments.
- (ii) 2-4 cm in length and covered by a thin epithelium.
- (iii) Stroma is divided into peripheral cortex and an inner medulla.

The above given features hold **true** for

- (1) Primary sex organ of females
- (2) Structural unit of excretory system
- (3) The structure that is the site of fertilisation
- (4) The structure forming birth canal.

139. Choose the **incorrect** match w.r.t. number of different structures present in *Periplaneta*.

- (1) Thoracic ganglia – 3
- (2) Abdominal ganglia – 4
- (3) Malpighian tubules – 100 to 150
- (4) Gastric caecae – 6 to 8

140. Read the given statements w.r.t. humans and select the **correct** option.

Statement A: Secretions of the male accessory glands constitute the seminal plasma which is rich in glucose, calcium and hormones.

Statement B: Secretions of prostate and seminal vesicle primarily help in the lubrication of penis.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

141. Observe the features mentioned below:

- (a) Streamlined body
- (b) Backwardly directed teeth
- (c) Presence of air bladder
- (d) External fertilisation

How many of the above mentioned features is/are common in both cartilaginous and bony fishes?

- (1) One
- (2) Two
- (3) Three
- (4) Four

142. "The diversity on the Earth was always the same since creation and will be the same in future."

The above statement holds **true** according to the

- (1) Theory of spontaneous generation
- (2) Theory of special creation
- (3) Theory of chemical evolution
- (4) Theory of panspermia

143. Listed below are some chemical compounds:

- (a) Water vapour (b) Carbon dioxide (c) Ammonia (d) Oxygen (e) Methane

Which of the above were released from molten mass that covered the early Earth's surface?

- (1) (a), (b), (c) and (d)
- (2) (b), (c), (d) and (e)
- (3) (a), (b), (c) and (e)
- (4) (a), (c), (d) and (e)

144. Consider the following statements w.r.t. humans:

- (a) Decrease in blood volume activates the osmoreceptors in our body.
- (b) ADH facilitates water reabsorption to prevent diuresis.
- (c) ADH can cause increase in blood pressure that leads to decrease in GFR.
- (d) Kidney function is efficiently monitored and regulated by hormonal feedback mechanism.

Choose the option representing only the **incorrect** statement(s).

- (1) (a), (b) and (c)
- (2) (b), (c) and (d)
- (3) (b) only
- (4) (c) only

145. Comb jellies are exclusively marine, _____ organisms with tissue level of body organisation.

Select the **correct** option to fill in the blank.

- (1) Triploblastic
- (2) Unicellular
- (3) Exclusively sessile
- (4) Radially symmetrical

146. Select the **correct** option w.r.t. the kidneys of a healthy adult human.

- (1) They are situated between the levels of 12th thoracic and last lumbar vertebra.
- (2) They are located close to the ventral inner wall of the abdominal cavity.
- (3) Each of them measures about 5 – 7 cm in length.
- (4) The thickness of each one of them is 2 – 3 cm.

147. One of the important objectives of biotechnology in the area of agriculture is to

- (1) Produce plants that are unable to tolerate abiotic stresses
- (2) Produce pest-resistant plants
- (3) Produce plants that are unable to efficiently utilise the nutrients present in the soil
- (4) Reduce the shelf life of plant products

148. Consider the following statements:

- (a) Fossils are usually the remains of soft parts of the life forms found in rocks.
 - (b) Fossils provide palaeontological evidences of evolution.
 - (c) Different-aged rock sediments contain fossils of different life forms.
 - (d) A study of fossils in different sedimentary layers indicates the geological period in which they existed.
- Select the option representing only the **correct** statements.

- (1) (a) and (b) only
- (2) (a), (b) and (c) only
- (3) (b), (c) and (d)
- (4) (b) and (c) only

149. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	<i>Ctenoplana</i>	(i)	Internal fertilisation
b.	Liver fluke	(ii)	Tissue level of organisation
c.	<i>Pennatula</i>	(iii)	Only sexual reproduction
d.	Sea lily	(iv)	Organ-system level of organisation

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iii), b(i), c(ii), d(iv)
- (3) a(i), b(iii), c(iv), d(ii)
- (4) a(iii), b(iv), c(i), d(ii)

150. A paleontologist has recently discovered a mosquito from more than 100 million years-old preserved amber. While extraction, most of the parts of the mosquito were lost, except for a small part of its leg. The researcher got curious and wanted to compare the DNA taken from the obtained sample with the DNA taken from the living mosquitoes. Thus, which of the following techniques will be most useful for him to increase the concentration of the mosquito's DNA?

- (1) Enzyme linked immunosorbent assay
- (2) Polymerase chain reaction by adding dNTPs such as uridylic acid in the reaction tube
- (3) Southern blotting followed by autoradiography
- (4) Polymerase chain reaction with the help of thermophilic polymerases

151. Match column I with column II and select the **correct** option w.r.t. blood cells and their functions.

	Column I		Column II
a.	Erythrocytes	(i)	Clotting of blood
b.	Neutrophils	(ii)	Associated with allergic reactions
c.	Eosinophils	(iii)	Perform phagocytosis
d.	Thrombocytes	(iv)	Transport respiratory gases

- (1) a(iv), b(iii), c(i), d(ii)
- (2) a(iv), b(i), c(ii), d(iii)
- (3) a(i), b(ii), c(iii), d(iv)
- (4) a(iv), b(iii), c(ii), d(i)

152. Choose the correct statement.

- (1) Gout is not a type of arthritis.
- (2) Muscular dystrophy is mostly due to genetic disorder.
- (3) Myasthenia gravis is a congenital defect which does not affect neuromuscular junction.
- (4) Decreased levels of gonadotrophins and increased levels of estrogen are common cause of osteoporosis

153. Choose the common property between muscles and neurons.

- (1) Excitability
- (2) Extensibility
- (3) Elasticity
- (4) Contractility

154.Assertion (A) : In a standard ECG, Q-wave is a negative wave.

Reason (R) : Q-wave represents initial phase of ventricular repolarisation

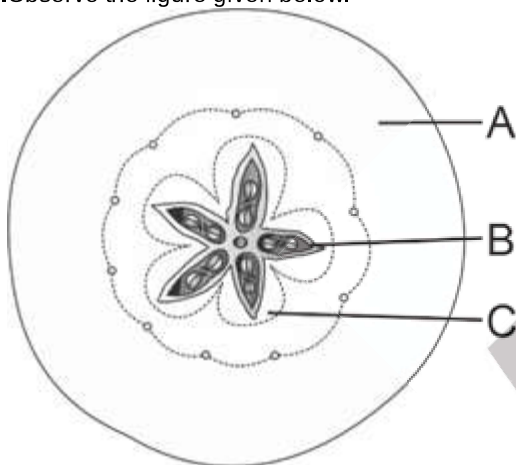
In the light of above statements, select the correct option

- (1) Both (A) and (R) are correct, (R) correctly explains (A)
- (2) Both (A) and (R) are correct, (R) does not explain (A) correctly
- (3) Both (A) and (R) are false
- (4) (A) is true, (R) is false

155._____ provide active while _____ provide passive immunity. Fill in the blanks respectively by selecting the correct option.

- (1) Antibodies given via ATS; Ready made antibodies received via the placenta from mother
- (2) Antibodies produced against weakened microbe that are administered through vaccination; Living microbes attacking our body
- (3) Anti-venom given after snake bite; Antibodies present in colostrum
- (4) Antibodies produced in body against microbial proteins; Anti-toxin administration in the body

156.Observe the figure given below.



Choose the **correct** option for A, B and C.

- (1) A-Thalamus, B-Mesocarp, C-Seed
- (2) A- Thalamus, B-Endocarp, C-Mesocarp
- (3) A-Pericarp, B-Seed, C-Endocarp
- (4) A-Epicarp, B-Mesocarp, C-Endocarp

157.Match the **column A** with **column B** and select the correct option.

Column-A	Column-B
a. Insect pollinated plant	i. Wheat
b. Wind pollinated plant	ii. Water hyacinth
c. Pollination takes place beneath the surface of water	iii. Vallisneria
d. Water pollination	iv. Zostera

- (1) a-ii, b-iii, c-iv, d-i
- (2) a-ii, b-i, c-iv, d-iii
- (3) a-iv, b-i, c-iii, d-ii
- (4) a-iii, b-i, c-ii, d-iv

158.Identify the **odd** one w.r.t. the characteristics of Down's syndrome.

- (1) Furrowed tongue
- (2) Partially open mouth
- (3) Congenital heart disease
- (4) Overall feminine development

159.Which of the following statements is **not** true?

- (1) Mendelian disorders are mainly determined by abnormal chromosomal arrangement
- (2) Sickle-cell anaemia is a qualitative problem of synthesising an incorrectly functioning globin
- (3) Possibility of a female becoming a haemophilic is extremely rare
- (4) A person affected with phenylketonuria lacks an enzyme that converts the amino acid phenylalanine into tyrosine.

160.A couple having a daughter with blood group A and a son with blood group AB. Which of the following could be the possible genotype of parents?

- (1) Father- $I^A i$, Mother- $I^B I^B$
- (2) Father- $I^A i$, Mother- $I^B i$
- (3) Father- $I^A I^A$, Mother- $I^B I^B$
- (4) Father- $I^A I^A$, Mother- ii

161.Read the Assertion(A) and Reason(R) statements given below and choose the **correct** option.

Assertion(A): The law of independent assortment does not hold true for the closely linked genes.

Reason(R): Closely located genes assort together.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

162.Non-vascular archegoniates are

- (1) Pteridophytes
- (2) Algae, fungi and lichens
- (3) Ferns
- (4) Liverworts and Mosses

163.Members of a class of algae in which cell wall is made up of cellulose and covered with gelatinous coating of algin is

- (1) *Fucus* and *Sargassum*
- (2) *Gelidium* and *Gracilaria*
- (3) *Dictyota* and *Chara*
- (4) *Ectocarpus* and *Polysiphonia*

164.Choose the **correct** option for the bioactive molecule streptokinase obtained from a bacterium.

- (1) It is used as blood cholesterol lowering agent
- (2) It is widely produced by bacterium *Streptomyces*
- (3) It is used to remove clot from blood vessels
- (4) It is used as an immunosuppressive agent in organ transplant patients

165.Commensalism can be exemplified by

- (1) Plover bird and crocodile
- (2) Lichens
- (3) Mycorrhizae
- (4) Barnacles on whale

166.Greater biological diversity is observed in tropical region due to all the given factors, **except**

- (1) Tropical latitudes have remained undisturbed for millions of years
- (2) Has less niche specialization
- (3) Higher productivity
- (4) They are less seasonal

167.Mark the **incorrect** statement.

- (1) Biological wealth of our planet is declining rapidly due to human activity
- (2) The degradation of many habitats by pollution also threatens the survival of many species
- (3) Many species have undergone extinction in the last 500 years due to overexploitation by human
- (4) The current species extinction rates are estimated to be 10 times slower than in prehuman era

168.Identify the statements as True (T) or False (F) w.r.t. *Ulothrix* and select the correct option.

- A. Both male and female gametes are non-flagellated and similar in size.
- B. It belongs to the class of algae which usually has rigid cell wall made up of an inner layer of cellulose and outer layer of algin.
- C. Most common spores produced are the zoospores.
- D. Major photosynthetic pigments are chlorophyll- *a* and *b*.

	A	B	C	D
(1)	T	F	T	T
(2)	T	T	F	F
(3)	F	F	T	T
(4)	F	F	F	F

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

169.Which one is **correct** with respect to primary treatment in a sewage treatment plant?

- (1) Flocs formation takes place during this step
- (2) Sedimentation occurs as a result of microbial oxidation
- (3) It is a physical process
- (4) This process involves degradation of organic and inorganic materials

170.In a field experiment when all the starfish were removed experimentally from an enclosed intertidal area of the American Pacific Coast

- (1) More than 10 species of invertebrates became extinct because of interspecific competition
- (2) The rate of speciation increased within a year due to intraspecific competition
- (3) All the vertebrates became extinct due to lack of food
- (4) Population density of invertebrates increased exponentially due to removal of important prey

171.The correct order of layers of adrenal cortex from inside to outside is

- (1) Zona glomerulosa → Zona fasciculata → Zona reticularis
- (2) Zona reticularis → Zona glomerulosa → Zona fasciculata
- (3) Zona fasciculata → Zona glomerulosa → Zona reticularis
- (4) Zona reticularis → Zona fasciculata → Zona glomerulosa

172. Complete the analogy w.r.t. the number of disulphide bonds present in a monomeric antibody molecule by selecting the **correct** option.

Intrachain disulphide bonds : 12 :: Interchain disulphide bonds : ____.

- (1) Two
- (2) Four
- (3) Six
- (4) Twelve

173. The type of immune response that is majorly responsible for graft rejection is

- (1) Mainly mediated by antibodies present in the blood
- (2) Provided by cells which assist B-cells in antibody production
- (3) A non-specific type of defence present since the time of birth
- (4) Mediated by cells which require only bone marrow for their maturation

174. Malarial parasites require two hosts for completion of their life cycle. Consider the following events w.r.t. their life cycle:

- (a) Sporozoites escape from gut and migrate to salivary glands.
 - (b) Parasites reproduce asexually in RBCs
 - (c) Fertilization of gametes occur in gut.
 - (d) Parasites reproduce asexually in hepatocytes.
- Select the option representing the **correct** sites of each of the above mentioned events.

- (1) Infected female *Anopheles* mosquito – a, b
Human – c, d
- (2) Infected female *Aedes* mosquito – a, c
Human – b, d
- (3) Infected female *Anopheles* mosquito – a, c
Human – b, d
- (4) Infected female *Aedes* mosquito – a, b
Human – c, d

175. How many of the following ARTs given in the box below involve *in vitro* fertilization?

AI, ICSI, IUI, GIFT, ZIFT

Choose the **correct** option.

- (1) Two
- (2) Three
- (3) Four
- (4) Five

176. For contraception, couples should abstain from coitus during which days in a normal 28 days menstrual cycle?

- (1) Day 1st to 5th
- (2) Day 10th to 17th
- (3) Day 21st to 27th
- (4) Day 5th to 12th

177. Proinsulin differs from mature insulin as former

- (1) Has more number of disulphide bonds in its structure
- (2) Lacks the C-peptide
- (3) Has only C-peptide with one intra chain H-bond
- (4) Is the inactive form of the latter

178. Which of the following is **incorrect** for the respiratory pigment present in erythrocytes in humans?

- (1) It represents the quaternary structure of protein.
- (2) O₂ binds with it in a reversible manner to form oxyhaemoglobin.
- (3) One molecule of it can carry a maximum four molecules of O₂
- (4) CO₂ and CO combine with it to form carboxyhaemoglobin and carbaminohaemoglobin respectively.

179. Read the following statements :

Statement A: An ideal contraceptive should not interfere with the libido of a person.

Statement B: Contraceptives are regular requirements for the maintenance of reproductive health.

Choose the **correct** option.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

180. Assertion (A) : If the levels of CO₂ increases in blood, then the rate of respiration decreases drastically.

Reason (R): Neural signals from the pneumotaxic centre can reduce the duration of inspiration when increased CO₂ concentration leads to low H⁺ level in the blood.

In the light of above statements choose the **correct** option.

- (1) Both (A) and (R) are correct, (R) correctly explains (A)
- (2) Both (A) and (R) are correct, (R) does not explain (A) correctly
- (3) Both (A) and (R) are false
- (4) (A) is true, (R) is false



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